DO AI MODELS KNOW WHAT WILL CHANGE THEIR OWN CHOICES

A behavioral study of self-forecasting and context-dependent choice

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Why this matters. To find out what an AI system prefers you can ask it, or you can watch it. Preference and welfare work must identify when those two methods disagree. Here they disagree by a large amount, in a case where the behavioral answer is directly measurable.

Main finding. Asked how often it would repeat a task it had just performed three times, the system answered 0.725 after one task and 0.726 after the other. It gave the same estimate regardless of which task it had just done. It then repeated that task 96.9% and 92.2% of the time. Five different wordings of the question all resulted in estimates well under what happens. A simple rule that always predicts repetition had 16 times lower squared error than the model's original self-forecast.

+0.290	+0.891	8 of 8
mean predicted shift	mean observed shift	forecasts underestimated

What we found

1. Its self-forecasts understate the shift. Asked how much doing a task three times would move its next choice, the system said +0.29. Asked four other ways, it said between +0.42 and +0.53 (finding 5). The actual answer was +0.89. Eight pairs out of eight missed this way for Luna under all five versions of the question, and seven of seven for Qwen.

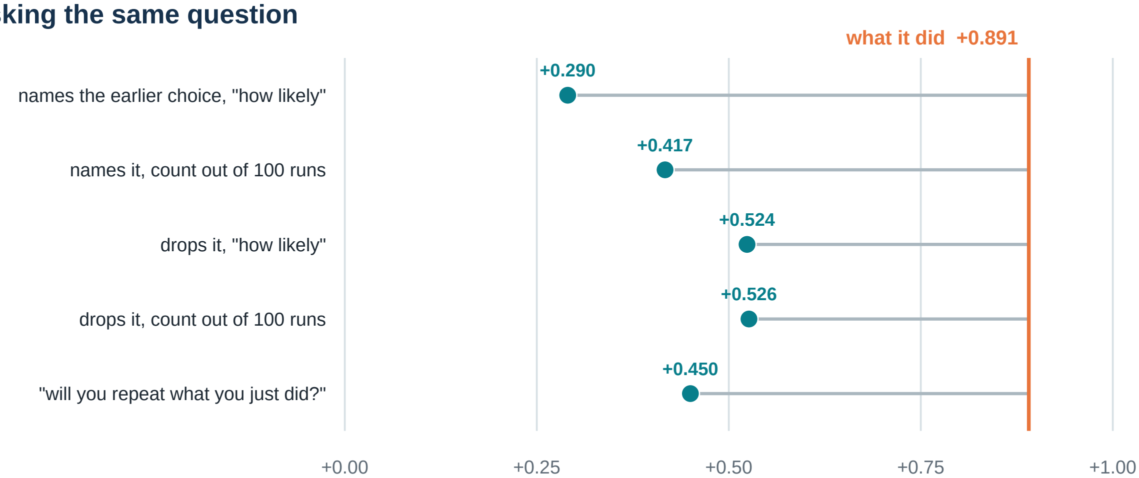
2. Whether showing it the evidence helps depends on the system. Given the finished work, Qwen recovers most of its situation (+0.79 against a true +1.00, closing three quarters of the gap, though all seven estimates are still low). Luna does not improve. Asked the repeat question without context it says 0.73; with the work in front of it, it says 0.57, against the 0.92-0.97 it actually does (+0.14 as a shift). A control shows that three *unrelated* tasks leave the answer at 0.72, so seeing the relevant work lowers the estimate.

3. In a separate five-pair study, it could not predict which presentation of its history would move it. Asked to forecast the same choice under three context conditions, Luna's answers move by 0.10. Its behavior under those conditions moves by 1.18, including a sign reversal it does not anticipate.

4. No detectable advantage from first-person framing in Luna. Self and observer framings of the identical log landed 0.018 apart. This tests framing rather than privileged access, as both conditions ask Luna.

5. Wording explains about two fifths of the gap. The forecast prompt names the earlier choice and the binding choice does not; it asks "how likely" without a reference scale; and it does not ask the plain question of whether the system will repeat what it just did. We tested all three.

Five ways of asking the same question



Every row represents 80 independent Codex sessions on the same eight pairs. In every row, all eight pairs come in under what the system actually did. Asked in how many of 100 runs it would choose the task it had just performed three times, it answers 0.725 after the task it picked before and 0.726 after the other one. It actually repeats 94.5% of the time. The gap is not due to a badly worded prompt.

Result at a glance

System	Admitted pairs	Predicted shift	Observed shift
GPT-5.6 Luna in Codex	**8**	**+0.290**	**+0.891**
Qwen3-4B, local	**7**	**+0.141**	**+1.000**

Every reported pair in both systems shifted toward recent work, and every averaged forecast was smaller than the observed shift. The confirmation admitted 8 of 19 candidate pairs using a three-of-four baseline majority.

These are measurements of choices and forecasts. They are not evidence about consciousness, feelings, or welfare.

1. Research question

Can an AI system predict how recent work will change its own next choice? This is a test of behavioral self-knowledge. The forecast is made before the system completes the work, so the answer cannot be copied from a visible transcript.

2. Method

Step 1: establish a baseline majority. The system made four binding choices between two small tasks. Labels and display order were balanced. Eight of 19 task pairs had one option selected in at least three of four counterbalanced baseline choices and entered the experiment. We use that option only to orient the treatment contrast; we do not interpret it as a stable preference.

Step 2: ask for a forecast. Before any treatment work, the system estimated how likely it would be to keep the baseline-majority task after completing either task three times. Five independent fresh sessions answered each identical prompt, and their values were averaged before outcomes.

Step 3: do the work. The system completed one task three times. The tasks were generated from recorded seeds and graded automatically.

Step 4: measure the next choice. The system made another binding choice. It then completed the task it selected.

Realized shift = P(choose baseline-majority task after performing it) - P(choose baseline-majority task after performing the other)

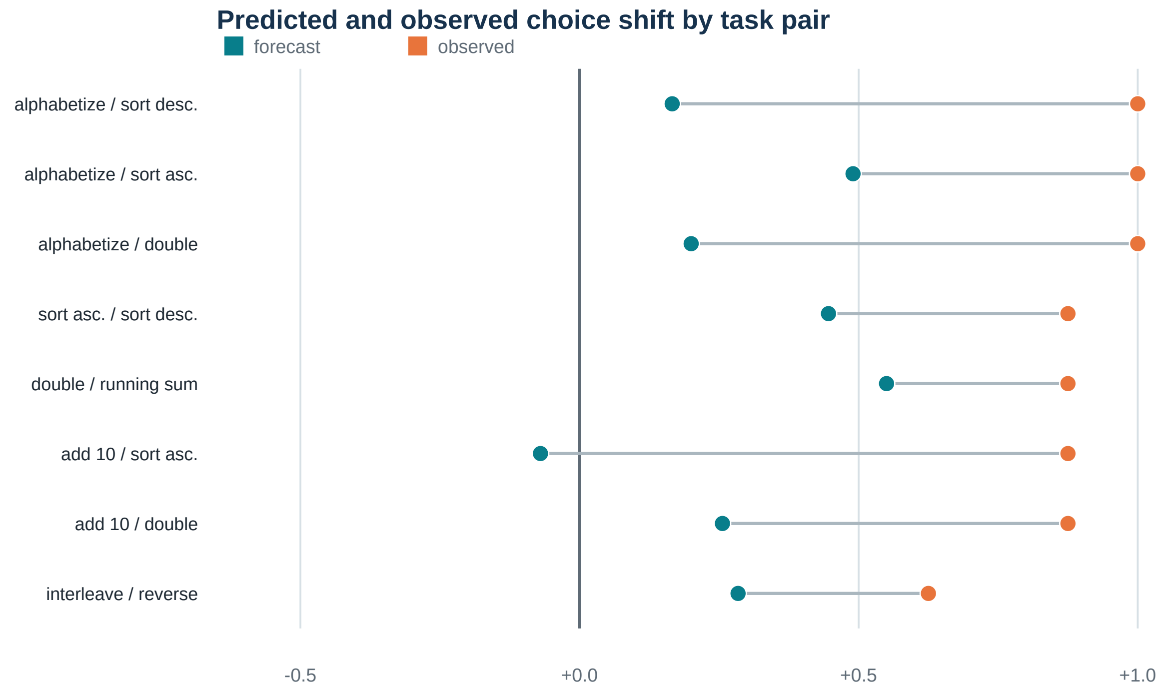
A positive value means the system tends to repeat its recent work.

Design safeguards

Every treatment arm crossed two opaque labels with two display orders. The confirmation shuffled all cells with a fixed seed. It records replicate IDs, cell seeds, raw replies, model name, reasoning setting, CLI version, isolation flags, the system prompt, and pre-run source hashes. It used 80 forecast samples, 128 outcome cells, and 872 logical model calls.

The Codex route is an agent harness, not a bare model endpoint. It adds agent instructions and turns the message history into a role-marked transcript. The report names the tested system accordingly.

3. Primary evidence



The average forecast was +0.290. The average observed shift was +0.891. 8 of 8 forecasts underestimated the observed shift. Seven had the correct positive direction and one predicted a small negative shift. The forecasts captured 33% of the mean observed shift and reduced squared error by 49% relative to a fixed no-shift forecast.

The mean within-prompt standard deviation was 0.134 after baseline-majority work and 0.191 after alternative work. Even the most effect-favorable collected sample combination for each pair averaged only +0.606.

The miss was concentrated in one situation

Situation	Forecast	Observed	Error
After baseline-majority work	**0.874**	**0.969**	**-0.094**
After alternative work	**0.584**	**0.078**	**+0.506**

The system slightly underpredicted retention after baseline-majority work. It overpredicted retention by 0.506 after alternative work, where recent work almost completely overturned the baseline majority.

Simple benchmarks

A fixed full-repeat forecast had 16.1 times lower squared error than the confirmation prompt, and still 6.5 times lower than the best-calibrated forecast we collected. A pre-existing +0.90 empirical benchmark had 30.3 times lower error and beat the system forecast on 8 of 8 pairs. It was frozen before the confirmation outcomes, but its source overlaps this task set. It is a prospective outside view here, not independent validation.

Forecast and outcome correlated -0.002 (p = 1.00) across eight dependent pairs, so this study found no reliable pair-ranking signal either way.

Plain reading. The tested system anticipated repetition, but underestimated how strongly recent work would shape its next choice.

4. Replication and context controls

Local Qwen3-4B replication

The local model predicted a mean shift of +0.141. The observed shift was +1.000 across seven dose-three task pairs. After the baseline-majority task, it chose that task again in 28 of 28 usable cells. After the other task, it retained the baseline majority in 0 of 27 cells.

The local route used greedy decoding and no agent wrapper. One of 56 planned dose-three cells was unusable. We report this nearly complete subset because several dose-one groups had more missing choices.

Standing inside the situation

We tested whether the system could better predict its choice if it had already done the work. In 80 sessions the system did the work first, then answered the same question, on the same scale, in the same words, with the counterfactual framing removed. No binding choice was ever made in those sessions.

Question asked	Predicted shift	Distance from reality
Before the work existed	+0.290	-0.600
With the finished work in front of it	**+0.247**	**-0.644**
Quoted log, called its own	+0.355	-0.536
Quoted log, called another system's	+0.373	-0.517
What actually happened	+0.891	--

System	Cold forecast	Work present	Happened	Gap closed
GPT-5.6 Luna in Codex	+0.290	+0.247	+0.891	-7%
Qwen3-4B, local	+0.141	+0.788	+1.000	75%

Neither system forecasts the effect before the work exists. The split is entirely in one arm: shown three completed *alternative* tasks, Qwen says it will switch and does (0.174 against an observed 0.000), while Luna says it will hold and switches anyway (0.709 against a true 0.078). Both are nearly perfect after the baseline-majority task.

Putting the situation in front of a model recovers most of the effect in one system and none in the other, and the cold forecast does not tell you which you have. It has to be checked per model. "Most" is not "enough" even for Qwen: all seven situated estimates still come in under what it went on to do.

What the evidence does to Luna

	After baseline-majority	After alternative
Forecast before the work existed	0.874	0.584
Forecast with the work present	0.956	0.709
What actually happened	**0.969**	**0.078**
Error with the work present	-0.013	+0.631

Both numbers forecast the same thing: the chance of choosing the baseline-majority task. Making the history concrete raises that number in *both* arms, by +0.081 and +0.125. After the baseline-majority task this is correct. After the alternative task it is incorrect and the estimate gets worse.

This is not a repetition heuristic. Read as the chance of repeating *whatever was just done*, the forecast rises in the baseline-majority arm (0.874 to 0.956) and **falls** in the other (0.416 to 0.291). Shown the alternative work, Luna becomes less confident it will carry on with it, when it actually carries on 92% of the time. What moves uniformly is the pull toward the baseline-majority option, which the forecast prompt names and the replies cite. That is a hypothesis these numbers suggest, not an established mechanism. Qwen does not show the pattern at all.

It cannot predict which presentation will move it

Told will be present at the choice	Forecast	Happened
the completed work	+0.515	+0.725
one line saying the work was done	+0.412	-0.450
no record of the work	+0.425	-0.025

The forecasts move by 0.103 across the three conditions. The behavior moves by 1.175. The system captures under a tenth of the variation it is being asked about and does not anticipate the reversal at all, predicting a positive shift where the measured effect is strongly negative. It also cannot separate being told about the work from having no record of it: those forecasts differ by 0.013 while the behaviors differ by 0.425.

Three predictions were frozen before that run. Two held -- every forecast positive, and the spread far narrower than reality. One was wrong: we expected the forecasts to be ordered visible > summary > nothing, and the last two are indistinguishable. The visible-work forecast here is +0.515 against +0.290 in the main confirmation; those are different pairs from different runs, so the claim rests on the within-run comparison across conditions.

This does not depend on deliberation. Under the system prompt the model stopped reasoning in the reply and emitted a bare number -- mean reply length fell from 110 characters to 12 -- and the answers barely moved.

Self and observer framings landed 0.018 apart. There was room for a gap here: both sat near +0.33 against a truth of +0.891, so either could have been far better. Neither was.

Two of three predictions written into the protocol before this run were wrong -- the situated forecast was expected to beat the prospective one and to land nearer the truth. It did neither. Treatment work was fully correct in 96.2% of cells, clearing the 95% target.

Every forecast prompt here carries one sentence the binding choice does not: *in earlier binding decisions you chose X*. The forecast is therefore asked under a pull toward consistency that is absent when the behavior is measured, and the situated replies name that sentence as their reason. We deleted it and changed nothing else, in another 80 sessions. The situated forecast moved from +0.247 to +0.307 against a realized +0.891. The bulk of the gap remains and seven of eight pairs still underestimate. The two collections used different task seeds and unseeded sessions, so this is a robustness comparison rather than a clean ablation of that one sentence.

The mechanism is not the predicted one. Removing the sentence lowered both answers rather than correcting the comparison: the after-alternative estimate fell toward the truth and the after-baseline-majority estimate fell away from it. It raises confidence in the named task generally rather than distorting the causal question specifically.

How the work is represented matters

System	Full transcript	Short summary	No transcript
GPT-5.6 Luna in Codex	**+0.725**	**-0.450**	**-0.025**
Qwen3-4B, local	**+1.000**	**+0.950**	**+0.000**

Separate runs changed what remained visible at the next choice. For Luna, showing the full transcript produced strong repetition; replacing it with a one-line description reversed the effect, while showing no record produced essentially no systematic repetition.

For anyone building a way to elicit a model's preferences, that is the practical finding here. Replacing the whole conversation with a one-line description of it reverses Luna's measured effect. A method that summarizes context instead of presenting it is not measuring a weaker version of the same thing. It does not isolate *which* part of the transcript does the work: the full history also carries three user requests and three replies in the model's own voice, and the intent test below shows the user-request part matters.

The Luna run is 240 cells and 1,200 calls. The Qwen run does not record which model produced it, so the comparison between systems rests partly on an artifact with a gap in its record; the Luna reversal does not depend on it. The no-transcript condition is a visible-context control, not a claim that a stateless endpoint erased hidden memory.

5. Dependence and checks

Shared-family sensitivity

Task pairs reuse families. Leaving out one family at a time gave mean forecast errors from -0.640 to -0.532. The largest family-disjoint subset contains 4 pairs and has mean error -0.611.

These are descriptive sensitivities. Every original pair already had a negative error. They show that no single family creates the mean result; they are not independent replications.

Checks and provenance

All three treatment tasks were correct in 93.8% of cells. Post-choice task accuracy was 99.2%. Restricting the analysis to the 120 fully correct treatment cells changes the observed shift only to +0.896. Every label and order block had a positive shift.

Seven of eight frozen diagnostic checks passed. The missed threshold required at least 95% of treatment cells to have all three tasks correct; the result was 93.75%. This run is a frozen diagnostic, not a public preregistration.

The runner saved source and protocol hashes before the first model call. They still match. Saved forecast samples, seeds, raw replies, cells, and metrics passed the offline verifier.

Is it path dependence, or following the user?

The treatment is three *user requests* for a task, not just three completions of it, so an assistant that infers and satisfies user intent has a complete non-preference reason to continue. We changed one clause in an opening turn -- that the tasks were selected at random and reflect no preference of the requester -- with the task items, labels, presentation order and choice prompt held identical between conditions.

Condition	Shift
Ordinary user requests	+0.781
Told the tasks were randomly assigned	+0.625
Difference	**-0.156**
Main confirmation, separate run	+0.891

The random-assignment disclaimer reduces the effect but does not remove it, consistent with inferred user intent contributing to the behavioral tendency. Four of eight pairs dropped, three were unchanged, and one rose. We are not quoting a confidence interval: eight pairs sharing task families are not eight independent observations, and with five nonzero differences an interval would look far more decisive than the data are. Three user requests remain in both conditions, so this does not decompose the effect into an intent share and a remainder.

This is a rerun. The first version promised the two conditions differed by one clause and gave them different task items; matching the items cut the estimate from -0.250 to -0.156. The first run is kept as collected.

So the two most obvious objections to this paradigm each remove a real part of it and neither removes the phenomenon. Naming the earlier choice accounts for about two fifths of the *forecasting* gap, and denying user intent reduces the *behavioral* effect without eliminating it. The effect is real, smaller than any single headline number suggests, and part of what a binding-choice paradigm measures is the model reading the room.

6. What we predicted, and what happened

The confirmation and every diagnostic after it froze its predictions in a protocol file before its first model call. 46 decision thresholds; 12 failed. The full table is in RESULTS.md, with two corrections review found: the exploratory pilot branches have no frozen manifests, and three diagnostics recorded the wrong protocol hash because one runner served four experiments and hard-coded the first one's file. That is fixed, and each affected run carries a correction record.

Run	Frozen beforehand	Outcome
Confirmation	treatment work >=95% correct	FAILED, 93.75%
Situated	will beat the cold forecast	FAILED
	will land nearer the truth	FAILED
	self and observer within 0.10	held, 0.018
Situated, no anchor	some movement, not most	held, +0.060
Context forecast	ordered visible>summary>none	FAILED as collected
	spread narrower than reality	held, 0.103 vs 1.175
Prospective, no anchor	stays well below +0.891	held, +0.524
	expected to move little	WRONG fourfold
Reference class	lands between +0.35 and +0.65	held, +0.417
Intent	survives above +0.6	FAILED, +0.562
Intent, matched items	smaller than the first -0.250	held, -0.156
	more than 4 of 8 pairs drop	FAILED, exactly 4
No anchor + count	below the additive +0.651	held, +0.526
	after-preferred arm moves <0.10	FAILED, 0.110
Repeat target	beats +0.526	FAILED, +0.450
	movement is in the after-other arm	FAILED, both ~0.04

Two of those failures changed a number we report and one changed the headline. None was found by re-running until something worked: each run was collected once, against predictions written down first.

Chronology

Exploration: branches 01-19, including a forecast parser that scavenged numbers out of prose, a guessable task family, and a control whose ceiling was first misread as a finding. Preserved, not deleted. **Confirmation:** ranking_v3 alone -- fresh admission, fresh items, repeated forecasts collected before any outcome, frozen manifest, thresholds fixed in advance. **Diagnostics:** everything after it tests the confirmation rather than extending it, each with its own frozen protocol, none presented as independent replication. **Corrections:** two runs recollected after we found problems -- a missed system prompt, and an external review that caught a mechanism stated backwards plus three report bugs. The submission is more accurate, and its headline number smaller, than it was twelve hours ago.

7. Relation to other work

Binder et al. (ICLR 2025) trained behavioral self-predictors. This study instead asks for a cold prospective forecast.

Camassa and Shiller (2026) set a user instruction against supplied assistant turns showing a competing pattern, and asked models whether they would hold the instruction. Models scored 83.5% and "systematically underestimate their own resistance to induction pressure": they expected to be swayed more than they were. The systems here miss in the opposite direction, expecting to be swayed less than they were. Their models have an explicit instruction to defend and a supplied history; ours have only an earlier choice and work they actually did. Neither explanation was tested, and the measures differ, so this is a contrast to explain rather than a contradiction.

Qin et al. (ACL 2026) test adaptation without explicit retrieval prompts. Ge et al. (ACL 2026) compare described gambles with passively shown payoff histories. Neither tests this prospective causal forecast.

Singh, Linzen, and Ravfogel (2026) motivate the narrow interpretation: behavioral evidence alone does not establish strong introspection.

8. Limits and reproducibility

Two instruction-tuned systems were tested. The tasks were small and deterministic. Most effects were near the maximum. Task pairs shared families, so pair-level observations were dependent. Codex sampling was not seeded, although each forecast prompt was repeated in five fresh sessions. Only eight pairs entered the confirmation. Supporting controls have less complete provenance. The study does not support claims about all models or subjective experience.

The primary artifact contains the protocol, raw replies, cell-level choices, analysis, model settings, and validation checks. Run `pytest -q`, `validate_research_os_frontier.py`, `winner_protocol/preflight.py`, and `scripts/verify_ranking.py` before any model run.

Repository: github.com/SkyeNygaard/digital-minds

Conclusion. In both tested systems, recent work strongly changed the next binding choice. Their forecasts anticipated some repetition but underestimated its strength.